

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

ME-705 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Operations Research and Supply Chain
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Solve the following problem using simplex method

$$\text{Maximize } Z = 10x_1 + 8x_2$$

$$\text{Subjected to } 2x_1 + x_2 \leq 40$$

$$2x_1 + 3x_2 \leq 80$$

$$\text{Where } x_1, x_2 \geq 0$$

2. Discuss and explain the MRP and ERP in brief.

3. Explain the terms :

i) Key decision

ii) Objective

iii) Constraints in the context of linear optimization models.

4. Write a note on difficulty in SCM due to uncertainty in demand.

5. What is logistics? Explain inbound and outbound logistics.

6. Discuss the impact of Bull-whip effect and outline how it leads to increased risk and poor service level. Comments on remedies to mitigate the impact of this phenomenon.

7. What is decision analysis? Explain it's importance.

OR

What is Meta-heuristics? Explain.

8. Write short notes on any two:

a) Safety stock and its importance

b) Que discipline in waiting line

c) Concept of competitive strategy

d) Assigning weight and consistency test of AHP
